Technical data
KEBALLOY ECO FE 191103 R-PET (Development product*)

KEBALLOY ECO FE 191103 PET is an electrically conductive PET compound with glass and carbon fiber reinforcement. The PET base polymer is made from post-consumer recycled PET bottles and the carbon fibers are post-industrial recycled. KEBALLOY ECO FE 191103 PET is thus a sustainable alternative to other electrically conductive structural materials.

Polymer: PET

ISO designation: PET-GF-CF(REC)

Productgroup: Electrically conductive compounds, Functionalized compounds

Brief description of the product family:

KEBALLOY ECO is our product family of engineering plastics and high-performance compounds based on post-consumer and post-industrial recyclates.

The portfolio enables manufacturers to integrate recycled content into technical applications while maintaining the performance characteristics required for their specific use case. Depending on the material selection and application, KEBALLOY ECO compounds can contribute to reduced product carbon footprint values compared to alternative virgin material solutions.

In addition, KEBALLOY ECO supports customer-specific recycling and take-back concepts, helping companies implement circular material strategies and make greater use of existing material resources.

Properties:

dimensionally stable, electrically conductive, good chemical resistance, good aging behavior, high continuous used temperature, High strength, High stiffness, semi-crystalline, PFAS-free

Typical areas of application:

Fasteners, Hardware, Housing, Sliding elements, Bearing bushes, Structural components

Industries:

Automotive, Railroad industry, Electrical and electronics industry, Industry, Agriculture, Mechanical Engineering, Furniture industry, Sports & Recreation

Physical properties

Density in kg/m³ | ISO 1183-1	1520

Mechanical properties

Tensile modulus in MPa | ISO 527-1	20000
Stress at break in MPa | ISO 527-1	160
Strain at break in % | ISO 527-1	1
Impact strength (Charpy) at 23°C in kJ/m² | ISO 179-1eU	26
Notched impact strength (Charpy) at 23°C in kJ/m² | ISO 179-1eA	9

Thermal properties

Melting temperature (DSC, 10°C/min) in °C | ISO 11357-1/-3	249
Burning behavior at thickness 0.8 mm | IEC 60695-11-10	HB

Processing instructions:

Pre-drying:

Dryer type: dry air dryer
Temperature: 80°C
Drying time: 4-8 h
Moisture content during processing: < 0.02%.

Recommended basic settings:

Melt temperature: 260 – 290°C
Mold temperature: 140 – 160 °C
Injection speed: high
Back pressure: 40 – 80bar (spec.)

Machine selection:

Screw: 3-zone screw with non-return valve
Nozzle: Open nozzle or shut-off nozzle
Wear protection: Wear protected according to machine manufacturer’s recommendation, suitable for processing fiber reinforced plastics
Injection unit: Shot volume = 50-80% of the maximum metering volume

Legal notices:

The information in this data sheet is based on our current knowledge and experience. Due to the wide range of possible influences during processing and application of our products, they do not exempt the processor from carrying out his own tests and trials. A legally binding assurance of certain properties or suitability for a specific application cannot be derived from our information.

* FE products are development products which are still in the trial phase. Technical data may still change in the course of product and process development. No final decision has yet been made on the commercialization of FE products. We reserve the right to discontinue the manufacture of FE products without giving further reasons.

Created at: 09.11.2026

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